Modelling and simulation of the energy system in Morocco in 2035

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Abstract— **This document explores the simulation and optimisation of the Moroccan energy system for 2035, providing a forward-looking analysis of the country's energy landscape. By estimating energy demand for that year, various simulation scenarios have been developed and examined in depth using EnergyPLAN software, enabling a comprehensive assessment of prospective energy strategies. Finally, an optimal energy mix and the ideal share of renewable energy for Morocco in 2035 were determined based on key factors such as cost, CO2 emissions, and energy imports and exports.**

Keywords— Energy policies; energy scenario; energy mix; MENA; Multidimensional Analysis; Renewable energy

I. INTRODUCTION

Morocco currently has around 41% of installed capacity from renewable energy (RE) and has an ambitious plan to boost the RE share to 52% by 2030. From previous scenarios in different countries, when RE share increases, energy storage and/or a strong connection with neighbouring countries are needed to compensate for the random nature of RE. Based on that, Morocco plans to develop its hydropower capacity, increase its interconnection capacity with Spain, and establish a hydrogen industry. The challenges that could face Morocco are as follows: 1) the risk of water and land requirements, 2) water requirements could lead to more water stress, 3) the requirements of large-scale RE installation (environmental, technical and economic factors), 4) building the hydrogen infrastructure, and 5) regulating measures, policies to regulate green hydrogen industry. Planning for future energy systems primarily relying on fluctuating RE sources, dynamic prices, and variable demands is a very strenuous problem. To design a sustainable and secure energy system with considerably high RE penetration levels, the security of supply and demand will face some difficulties due to the uncertainties in the RE forecasting data over more extended periods. The research questions are: What is the optimum energy mix? How much dispatchable power, storage and flexible demand is needed? The impact of uncertainties? The proper Energy transition plan? A detailed analysis of the current energy system, the available renewable energy resources, and the anticipated evolution of the demand should be carried out to answer these questions. In this paper, we have simulated the optimum energy mix for Morocco in 2035. The paper is organised as follows: in section II, we present the data used in the analysis and explain the method used to carry out the study. In section III, we analyse the different future energy scenarios to find the optimum energy mix for Morocco in 2035, and finally, the drawn conclusions are given in section IV.

II. DATA AND METHODOLOGY

Morocco has a total installed generating capacity of about 11 GW and 4.03 GW of renewable energy sources [1]. Morocco plans to increase its renewable energy share of produced electricity to 52% by 2030, from which 20% solar, 20% wind, and 12% hydro [1]. Due to Morocco's location, Morocco has a high solar irradiation, which significantly facilitates the implementation of solar power generation. Figure 1 below shows the normal direct irradiation, and Figure 2 shows the wind energy map. Regarding wind turbine electricity generation capacity, Figure 2 shows how Morocco has different locations with the potential to locate large wind farms. The most significant area is along the country's Atlantic coast, particularly in the southern region, where wind speeds regularly surpass 8 m/s. The anticipated maximum theoretical potential of wind power in Morocco is 25 GW [1].

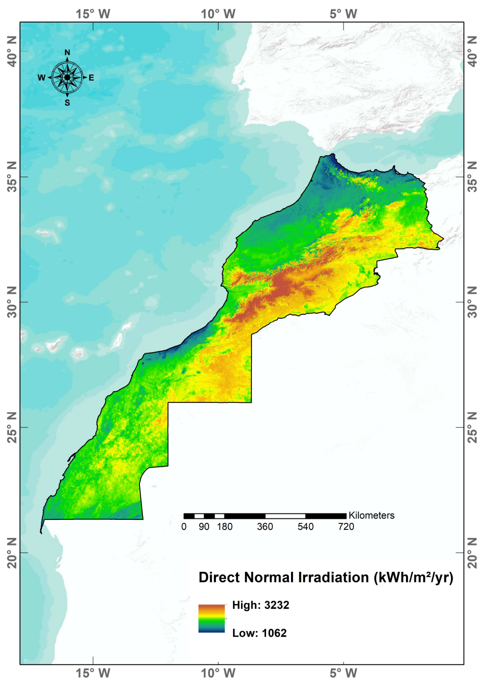

Fig. 1. Solar maps of Morocco. DNI [kWh/year/m2]. [2-4]

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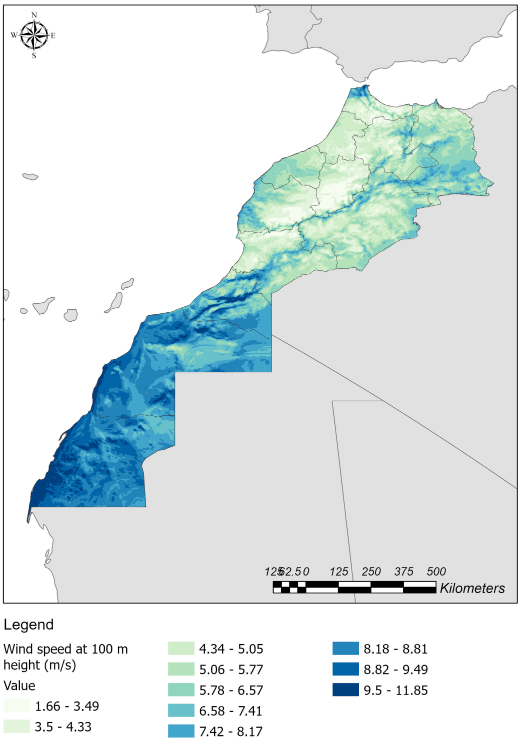

Fig. 2. Average wind speed at 100 height [m/s] (source: GEP/Vortex) [5].

Large PV systems installation is a viable solution for Morocco to break the dependence on fossil fuel as the levelized cost of electricity produced from PV is around 0.06 \$/kWh [6], and the production cost for one kWh produced from wind is 0.038 \$ [7]. The authors in [8] investigated Morocco's energy scenario in 2030, and they conclude that Morocco should take more measures to achieve its renewable energy goals, which include grid modernisation, efficiency improvements, strengthening interconnections with neighbouring countries, and expanding the storage capacity.

Simulation software is used to analyse the energy mix in Morocco. EnergyPLAN simulates the national energy system at hourly intervals, including not only the electricity part of the country but also sectors such as heating, cooling, industry and transport [8]. The data needed for EnergyPLAN are energy consumption, solar energy and wind energy resources. The anticipated energy consumption in Morocco is shown in Table 1 [9],[10]. Figure 3 shows the scheme of the energy system used by EnergyPLAN to simulate the different energy scenarios, with the various technologies used and the energy flows.

TABLE I. ENERGY CONSUMPTION IN 2035

Low-Growth scenario (2.6%) [TWh]	Base scenario (3.9%) [TWh]	High-Growth scenario (5.6%) [TWh]
56.85	70.23	90.61

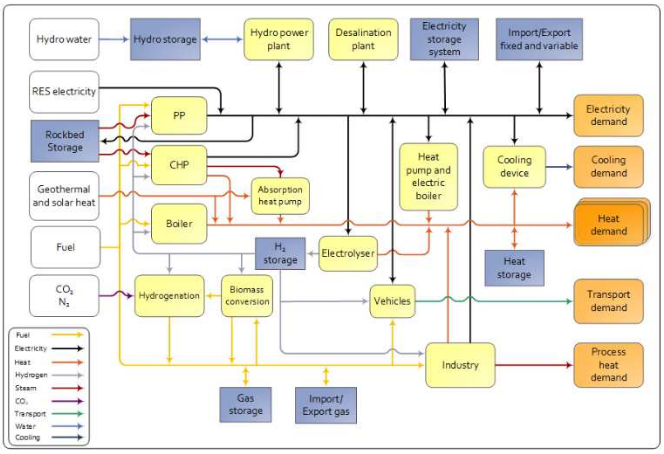

Fig. 3. Scheme of the technology and flow of EnergyPLAN

III. SCENARIOS FOR MOROCCO IN 2035

The scenarios to be investigated are defined by the anticipated energy consumption in the coming years based on different growth rates for Morocco and according to the generation technologies and the various energy storage technologies to be implemented. Eighteen scenarios have been simulated, as given in Table II. For each scenario, thousands of sub-scenarios are simulated with varying energy mixes. To carry out this simulation, the MATLAB Toolbox for EnergyPLAN created by Cabrera et al. [11] will be used, making it easier and faster to study the different cases and find the most optimal solution for each scenario. These values have been chosen so that the mid values selected (the third value of the five for each of the renewables used as variables), and taking into account the installed capacity of the non-renewables, generate the situation of the percentage of renewable capacity in each scenario. More clearly, for a scenario characterised by an installed renewable capacity of 59%, the values that achieve (approximately) this situation are the third, added to the fixed renewables (dammed hydro and river hydro). In this way, we ensure that we analyse situations above and below the average for every one of the scenarios.

For each scenario defined above, 625 combinations of capacities of the different variables defined have been studied, corresponding, as already mentioned, to wind power (on and off-shore) and solar power (PV and CSP). When the different scenarios were investigated separately, we analysed the optimal situation for each scenario. The output parameters to be studied are:

- **CO_2 emissions, in Mt/year**: Output showing CO_2 emissions from electricity generation by fossil fuel-based generation systems. The aim is to minimise this output.

- **Imports, in M€/year:** Value reflecting annual purchase of electricity from foreign countries. The aim is to minimise this output.

- **Exports, in M€/year:** Value reflecting the annual sale of electricity to foreign countries. The aim is to minimise this output (It is intended to mitigate because the value provided by the EnergyPLAN software is negative, representing a "cost reduction." Therefore, the lower the obtained value, the greater the electricity export).

- **Total costs, in M€/year:** This output considers all factors related to the costs of the electricity system, including all electricity generation and storage systems used (maintenance costs, annual amortisation, etc., depending on the capacity of each of these), the import-export balance, electricity connections with the foreign countries, etc. The aim is to minimise this output.

TABLE II. ENERGY SCENARIOS IN 2035

SCENARIO	RES [%]	CONSUMPTION [TWh/year]
A	100	56.85
B	59	
C	69.62	
D	100	70.23
E	59	
F	69.62	
G	100	90.61
H	59	
I	69.62	
J	100	56.85
K	59	
L	69.62	
M	100	70.23
N	59	
O	69.62	
P	100	90.61
Q	59	
R	69.62	

Finally, when trying to find the most optimal situation (both for the individual study of each scenario and for the scenarios as a whole), results will be obtained depending on whether the most economical scenario is desired (more weight will be given to the output "Total costs"), the most ecological (more weight will be given to the production "CO2 emissions"), the one with the most electricity aid to Europe (more weight will be given to the production "Exports") and a “normal” one (all the outputs with the same weight), always seeking a balance between all the outputs. To achieve this, a multivariable optimisation will be performed, and using the studied outputs, the aim is to find the combination of installed renewable capacity that yields the most favourable scenarios for the desired objectives. The function used is of the form:

$$f(x_1,x_2,x_3,x_4)=a\,x_1 + b\,x_2 + c\,x_3 + 1/(d\,x_4) \quad (1)$$

Where x_1: “Total Costs” variable, x_2: “CO2 emissions” variable, x_3: “Import” variable, x_4: “Export” variable, a, b, c, d: Weights assigned to each variable according to the case study to be analysed. The weighted sum multi-objective optimises this function because this method combines the multi-objective functions into one scalar. The main problem with this method is related to the weights given to each variable through the different coefficients since depending on the value it is decided to assign to each one will directly and notably influence the solution obtained [12]. Despite that, this optimisation method has been chosen because, thanks to the different values that the weights can get, different optimisation situations can be generated. To obtain the optimal situation for each case study, the goal is to minimise the variables "Total Costs," "CO2 emissions," and "Import" and maximise the variable "Export." Therefore, the objective will minimise function $f(x_1,x_2,x_3,x_4)$. The values of the constants for the different cases are shown in Table III. Figure 4 clearly shows the cases that have proved optimal for the various scenarios. The number in each box represents the optimal 625 situations analysed for each scenario, with 1 having the lowest installed capacity of the four renewables and 625 with the highest.

TABLE III. VALUES OF THE CONSTANTS ACCORDING TO THE CASE STUDY

Study case	***a***	***b***	***c***	***d***
Economic	10	1	1	1
Ecologic	1	10	1	1
Export	1	1	1	10
Normal	1	1	1	1

Scenario	wer TotalCos	Lower CO2	Max Exports	Opt. Econ.	Opt. Ecol.	Opt. Exp.	Opt. Norm.	RES Cap.	Cons.	Stor.
A	605	-	625	625	625	625	625	100	57	Short
B	601	620	625	625	625	625	625	59	57	Short
C	601	620	625	625	625	625	625	70	57	Short
D	605	-	625	625	625	625	625	100	70	Short
E	601	625	625	625	625	625	625	59	70	Short
F	601	625	625	625	625	625	625	70	70	Short
G	605	-	625	625	625	625	625	100	91	Short
H	601	625	625	625	625	625	625	59	91	Short
I	601	620	625	625	625	625	625	70	91	Short
J	605	-	625	625	625	625	625	100	57	Long
K	601	620	625	625	625	625	625	59	57	Long
L	601	620	625	625	625	625	625	70	57	Long
M	605	-	625	625	625	625	625	100	70	Long
N	601	620	625	625	625	625	625	59	70	Long
O	601	625	625	625	625	625	625	70	70	Long
P	605	-	625	625	625	625	625	100	91	Long
Q	601	625	625	625	625	625	625	59	91	Long
R	601	620	625	625	625	625	625	70	91	Long
ALL	601 (L)	620 (L)	625 (L)	625 (L)	625 (L)	625 (L)	625 (L)	-	-	-

Fig. 4. Most optimal results for each scenario and case analysed

Scenario L situation 601 is the lower total costs situation. The results of this scenario are shown in Table IV; the analysis of scenario 601 of Scenario L is shown in Figure 5, which corresponds to the situation with the lowest total costs of all the scenarios analysed. This scenario corresponds to the following values: 4500 MW of wind, 2500 MW of photovoltaic and 0 MW for both CSP and offshore wind. Despite this high percentage of installed capacity, it can be seen that the total electricity generated by renewables is 23 TWh per year, which would represent 40.4% of the total electricity generated. This can also be seen in the annual average of RES used, which stands at 2402 MW, while non-renewable capacity is 3797 MW. The second optimum scenario is 620, with the installed renewable capacities of 4500 MW of wind, 2500 MW of photovoltaic, 2227.5 MW of CSP and 4000 MW of offshore wind. The results of this scenario are shown in Table V. The results of this situation are shown in Figure 6. The installed capacity of renewables already reaches 72.15% of the total, which is very close to the value established for this scenario. This high level of installed capacity is reflected in the electricity generated by the renewable energy group. The annual generation obtained from renewable sources is 35.3 TWh, accounting for 62% of the total electricity generated. This superiority is also shown in the average yearly production of the two groups: 4025 MW from RES and 2768 MW from non-renewable sources. The third optimum scenario is 625, which corresponds to the installed capacities of the variables corresponding to renewable energies, which are 4500 MW of wind, 2500 MW of photovoltaic, 2970 MW of CSP and 4000 MW of offshore wind. The results of scenario 625 L are shown in Table VI. It should be pointed out that this situation corresponds, as is evident, to the one with the highest installed capacity corresponding to renewables with a rate of 72.99%, placing it slightly above that established for Scenario L.

TABLE IV. MAIN RESULTS AND DATA OF SITUATION 601 IN SCENARIO L (LOW-COST ELECTRICITY CASE)

Parameter	Value	Unit
Actual RES capacity	54.39	%
Total annual costs	55995	M€
CO_2 emissions	23	Mt
Electricity import	63	M€
Electricity export	16	M€
Electricity from RES	23	TWh
RES share (primary energy)	23.7	%
RES share (electricity)	40.4	%
Total electricity produced	56.93	TWh
Average RES production	2617	MW
Average Non-RES prod.	3797	MW

TABLE V. MAIN RESULTS AND DATA OF SITUATION 620 IN SCENARIO L (LOW-COST ELECTRICITY CASE)

Parameter	Value	Unit
Actual RES capacity	72.15	%
Total annual costs	56689	M€
CO_2 emissions	14.97	Mt
Electricity import	46	M€
Electricity export	293	M€
Electricity from RES	35.3	TWh
RES share (primary energy)	39.5	%
RES share (electricity)	62	%
Total electricity produced	56.93	TWh
Average RES production	4025	MW
Average Non-RES prod.	2768	MW

TABLE VI. MAIN RESULTS AND DATA OF SITUATION 625 IN SCENARIO L (LOW-COST ELECTRICITY CASE)

Parameter	Value	Unit
Actual RES capacity	72.99	%
Total annual costs	56871	M€
CO_2 emissions	14.07	Mt
Electricity import	46	M€
Electricity export	358	M€
Electricity from RES	36.6	TWh
RES share (primary energy)	41	%
RES share (electricity)	64.3	%
Total electricity produced	56.92	TWh
Average RES production	4180	MW
Average Non-RES prod.	2703	MW

In terms of electricity generation, in this case, 36.6 TWh of electricity comes from renewable sources, which represents 64.3% of the total generated. Once again, this superiority can be seen in the average production of renewable and non-renewable sources, with values of 4180 MW and 2703 MW, respectively. Based on the above, it is concluded that the present scenario would also correspond to the lowest CO2 emissions (14.07 Mt scenario 625 of Scenario L, 14.97 Mt scenario 620 of Scenario L). Finally, the scenarios are compared in Figure 8—a direct comparison of installed renewable capacities between the early 2020s and the most optimal scenarios for 2035. As mentioned, this shows the significant increase in installed capacity required for all the technologies analysed to reach these scenarios, which are the most favourable for Morocco's electricity future.

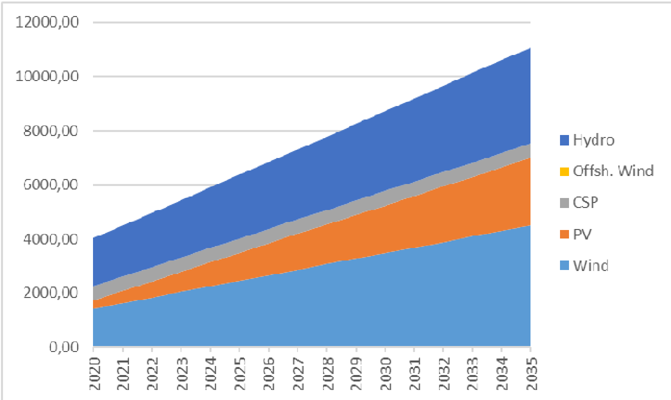

Fig. 5. RES installed capacity increase for Scenario L situation 601

Something that has been mentioned, and which should be emphasised importantly, is the economic aspect. Although the optimum results obtained indeed are those that have been shown, a significant financial effort will be required on the part of Morocco to make this leap towards energy independence in terms of imports and as a country providing electricity support for the European continent. For this reason, conducting an economic feasibility study of these projects is necessary so that, if it proves impossible to carry them out, a similar energy situation should be found that meets the economic requirements.

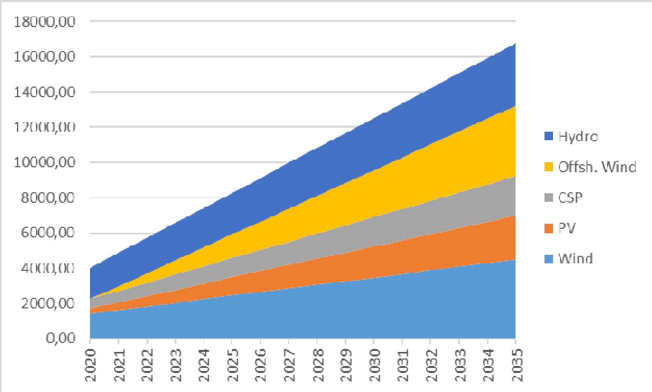

Fig. 6. RES installed capacity increase for Scenario L situation 620 [MW]

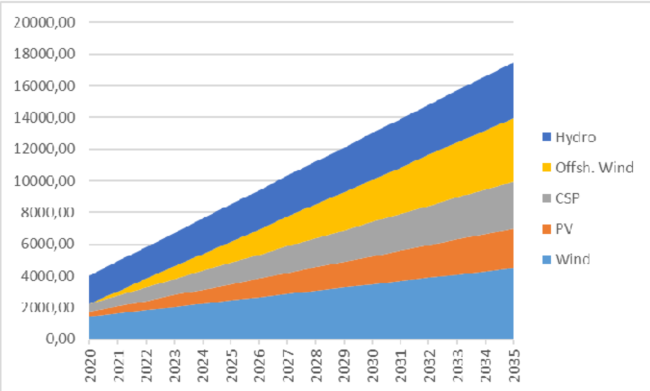

Fig. 7. RES installed capacity increase for Scenario L situation 625 [MW]

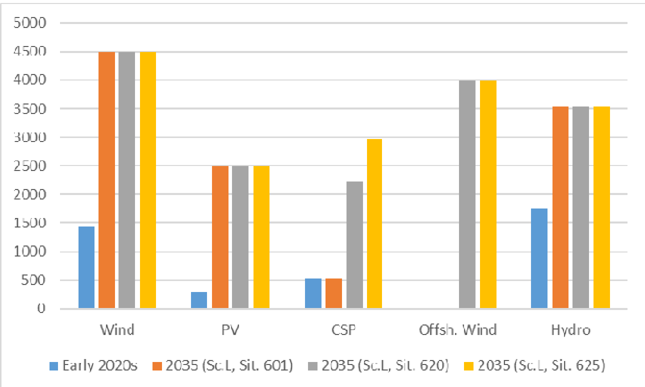

Fig. 8. RES installed capacity [MW]

IV. Conclusions

Once the simulations of all the situations of the different scenarios and the in-depth analyses of those that had been more favourable according to the different factors that had previously been established as determining factors had been carried out, it was concluded that Scenario L, and more specifically, situation 625, which has the largest installed capacity of renewable energies, was the most favourable. This scenario has an installed renewable capacity of around 69.62% and long-term electricity storage. In addition, this is within the electricity generation ranges established in the country's energy analysis, so it can be considered a satisfactory result. On the other hand, leaving aside the differences in electricity consumption between the different scenarios, it has been possible to observe a constant among the results that minimise (or maximise, in the case of exports) the factors analysed, and that is that in (almost) all of them, the most optimal scenarios also correspond, as in the case analysed for all the situations, with those that have around 69.62% of installed renewable capacity and long-term electricity storage.

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